State-Dependent Option Market Repricing Around Earnings Announcements

Thesis Presentation

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Outline

1. Introduction
2. Literature and Intuition
3. Data and Variable Construction
4. Main Results
5. Robustness and Interpretation
6. Main Findings Summary
7. Contributions and Implications
8. Conclusion
9. Thank You
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A Quick Review of Our Main Topic

- Earnings announcements are recurring, firm-specific, economically important events.
- Prior literature: announcements reduce implied volatility (IV) $\rightarrow$ "volatility crush".
- **This paper adds two overlooked dimensions:**

Dimension	Question
Structure (skew)	Does implied volatility skew also normalize after the announcement?
State (VIX)	Does aggregate market fear (VIX) weaken the announcement's effect?

Two main findings (preview):

1. Steeper pre-announcement skew $\rightarrow$ stronger post-announcement skew normalization (coefficient ≈ 0.63).
2. Higher pre-announcement VIX $\rightarrow$ weaker and delayed IV compression.

These findings build on three strands of literature. Let us briefly revisit them.

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Literature Review – Three Pillars

Literature	Core Finding	Gap / What this paper adds
1. Disclosure & uncertainty resolution (Kim & Verrecchia 1991/1994; Patell & Wolfson 1981)	IV drops after earnings → uncertainty resolved	Focuses only on IV level, ignores skew (asymmetric tail beliefs)
2. Option skew & tail beliefs (Bali & Murray 2013, etc.)	Skew reflects downside fear; predicts crash risk	Skew used as a *predictor*, not as an object to be *corrected* by disclosure
3. Macro state dependence (Pastor & Veronesi 2012, 2013; Baker, Bloom & Davis 2016)	High macro uncertainty → risk premia rise, firm-specific info matters less.	No systematic study of how VIX modulates earnings-announcement repricing

Research Gap

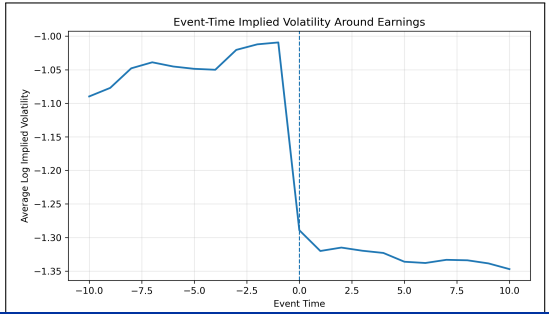
- Does earnings news *correct/normalize* pre-event skew?
- Does high VIX *weaken or delay* uncertainty resolution?

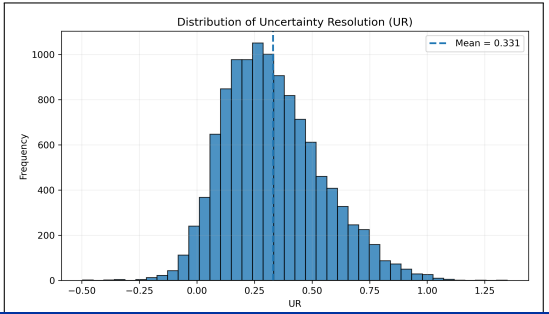
Given this background, we first confirm the known volatility crush in our own sample.

What We Already Know – The Volatility Crush

- Replicate the standard result using our sample (2015–2025, $\sim$ 11,468 firm-quarter events).
- **Average UR (Uncertainty Resolution) is positive** $\rightarrow$ IV declines after earnings.

Variable	Mean	Interpretation
$UR = -(IV_{+1} - IV_{-1})$	≈ 0.33	Positive $\rightarrow$ IV drops on average (Positive values indicate uncertainty resolution/skew normalization
$SR = -(Skew_{+1} - Skew_{-1})$	≈ -0.02	Close to zero $\rightarrow$ skew does not systematically change on average





The Missing Piece – Implied Volatility Skew

Skew definition:

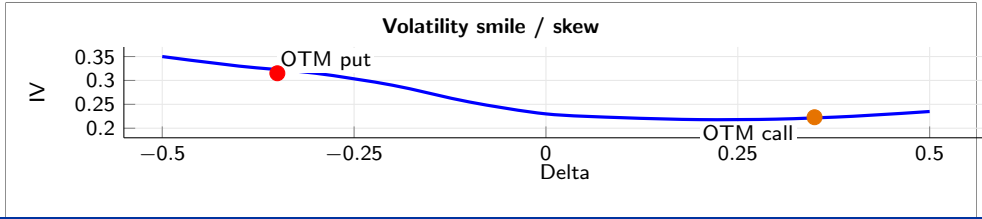
$$Skew_{i,t} = \ln\left(\frac{IV_{OTM}^{put}}{IV_{OTM}^{call}}\right)$$

- Positive value → put is more expensive than call → investors fear downside.

This paper's new variables:

Variable	Definition	Interpretation
PreSkew	Skew measured on day -1	Pre-announcement downside fear
SR (Skew Revision)	$-(Skew_{+1} - Skew_{-1})$	Positive SR = skew declines after announcement

Key question: Does a higher *PreSkew* lead to a larger *SR*?
⇒ That would mean earnings announcements correct exaggerated downside pricing.



Hypothesis 1 – Skew Normalization

H1: Higher pre-announcement skew (PreSkew) $\rightarrow$ stronger post-announcement skew revision (SR).

Intuition:

- Steep skew = overpriced downside insurance due to fear of bad news.
- Earnings release resolves firm-specific uncertainty $\rightarrow$ demand for puts falls $\rightarrow$ skew declines.

Alternative explanation (to rule out):

- Skew simply mean-reverts regardless of announcements.

What this hypothesis implies:

Dimension	Implication
Market efficiency	Systematic skew reversal suggests that downside-tail pricing may become temporarily stretched before disclosure.
Role of disclosure	Mandatory earnings reports have a corrective function – reduce predictable tail-pricing distortions

But even if skew normalizes, the overall uncertainty resolution might depend on the macro environment – especially the VIX.

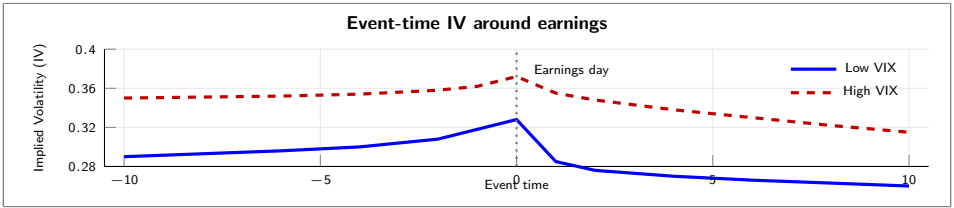
The Role of Macro State (VIX)

- Investors interpret firm-specific news against the aggregate risk backdrop.
- **Two plausible mechanisms for high-VIX states:**

Mechanism	Explanation
Attention-based	Investors focus on macro shocks, pay less attention to firm-level earnings news.
Risk-premium based	Even if firm risk falls, macro risk premia keep IV elevated.

Define second outcome variable: $UR = -(IV_{+1} - IV_{-1})$

- Positive UR = IV declines.
- Expectation: high pre-announcement VIX → lower UR.

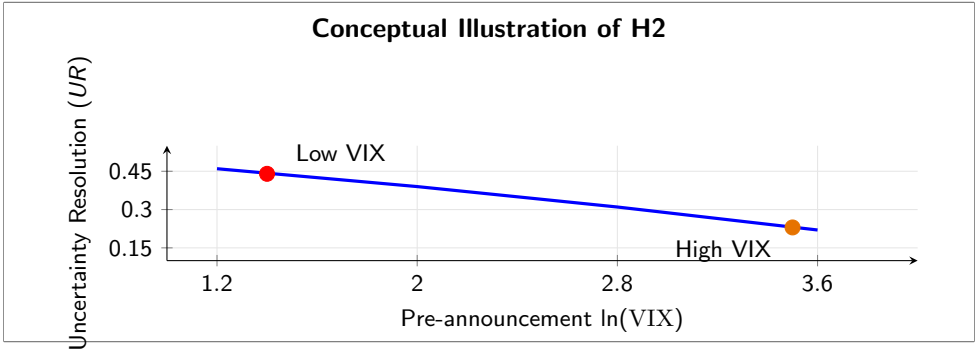


Hypothesis 2 – High VIX Dampens Uncertainty Resolution

H2: Higher pre-announcement VIX $\rightarrow$ weaker post-announcement IV compression (lower UR).

- Preview of richer finding: later we show that high VIX *delays* rather than eliminates resolution.

H2 prediction: coefficient on
$LN_VIX < 0$



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Data & Sample Construction

Item	Description
Sources	I/B/E/S (dates), OptionMetrics (options), CRSP (stocks), Compustat (accounting), CBOE (VIX)
Sample period	2015–2025
Final observations	$\approx$ 11,468 firm-quarter events (after matching pre- and post-announcement data)
Option filters	Maturity 10–60 days; select OTM put/call with delta closest to 0.25 (between 0.1 to 0.6)
Event window (main)	$[-1, +1]$ trading days (narrow window to capture immediate repricing)

The key variables and summary statistics are shown on the next slide.

Variable Definitions & Summary Statistics

Variable table (simplified):

Variable	Definition	Role
PreSkew	Skew on day -1	H1 independent variable
SR	$-(Skew_{+1} - Skew_{-1})$	H1 dependent variable
LN_VIX	$\ln(VIX)$ before announcement	H2 independent variable
UR	$-(IV_{+1} - IV_{-1})$	H2 dependent variable
PreIV, size, volume, return, etc.	Controls	Covariates

Summary statistics highlights (from paper's Tables 1 and 2):

- UR mean ≈ 0.33 (positive, volatility crush confirmed)
- SR mean ≈ -0.02 (skew does not automatically normalize on average)
- **Low-VIX states:** mean UR ≈ 0.373
- **High-VIX states:** mean UR $\approx 0.288 \rightarrow$ raw difference supports H2

With these summary patterns in mind, we now turn to the regression evidence, starting with Hypothesis 1.

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Result 1: Pre-Announcement Skew and Skew Normalization

Specification	PreSkew Coeff.	t-stat	Adj. R^2
Baseline	0.6325	53.69	0.306
+ Rich Controls	0.6315	53.29	0.312
+ Absolute Return	0.6315	53.23	0.312

Dependent variable: SR from day −1 to day +1.
$N \approx 11{,}468$.

Key Takeaways

- Positive, large, and stable coefficient across specifications
- Steeper pre-announcement skew predicts stronger normalization
- **Supports H1:** announcements correct downside-tail pricing more strongly when pre-event fear is elevated

Economic Magnitude

$$0.632 \times 0.065 \approx 0.041$$

Takeaway

Pre-announcement downside fear directly determines the intensity of pricing correction during earnings events.

Result 2: High-VIX States Weaken Immediate Resolution

Dependent Variable	Coef. on LN_VIX_pre	t-stat
UR (Main)	-0.155^{***}	-10.32
UR (Rich Controls)	-0.173^{***}	-14.29
UR / PreIV	-0.333^{***}	-10.11
IV post	0.174^{***}	14.36

Firm-clustered standard errors. *** $p < 0.01$.

What this means

- Higher VIX predicts weaker post-earnings volatility compression
- Pattern is robust to alternative outcome definitions and controls

Why column (4) matters

- In high-VIX states, IV not only falls less, but remains at a higher level after the event

Takeaway

When aggregate market fear is elevated, earnings announcements resolve less immediate option-implied uncertainty.

Timing Version of H2: Immediate vs. Delayed Repricing

Window	Coef. on LN_VIX_pre	t-stat
UR1 ($-1 \to +1$)	-0.169^{***}	-14.63
UR23 ($+2 \to +3$)	0.122^{***}	12.99
UR45 ($+4 \to +5$)	0.076^{***}	7.94

Segmented windows. *** $p < 0.01$.

Key Findings: From Dampening to Catch-up

- **Initial Dampening** (UR1): In high-VIX regimes, the immediate Vol Crush is significantly muted.
- **Delayed Resolution**(UR23, UR45): The coefficient sign inversion confirms the repricing is **postponed**, not eliminated

Interpretation: Signal-to-Noise Ratio (SNR)

- **Macro Noise vs. Firm Signal**: VIX represents aggregate "noise" that obscures firm-specific earnings "signals".
- **Information Processing Friction**: This pattern is **consistent with** models of limited attention or slower Bayesian updating under macro-economic stress.
- **Pricing Inefficiency**: The findings **point toward** temporary challenges to immediate informational efficiency, rather than a permanent market failure.

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Robustness Checks

H1 Robustness

- Alt windows: results remain positive $[-2, +2] \to 0.593$; $[-1, +5] \to 0.682$
- Fama–MacBeth: 0.652 ($t = 23.13$)
- Portfolio sorts: monotonic increase in SR across PreSkew deciles

H2 Robustness

- Normalized UR: result holds ($\beta = -0.333$)
- Post-IV level: positively related to pre-VIX ($\beta = 0.174$)
- Fama–MacBeth: consistent sign, though weaker statistical power

Summary

H1 is especially robust across methods; H2 remains strong in pooled event-level regressions and alternative outcomes, though weaker in Fama–MacBeth.

Interpretation and Identification

Leakage Concerns

CONCERN

- Pre-skew reflects private information.

ACTION

- Narrow event windows
- Industry/Quarter FEs
- Control for analyst disagreement

Mean Reversion

CONCERN

- Skew declines mechanically

INTERPRETATION

- Coefficient stability suggests results are not trivially driven by reversion

General Persistence

CONCERN

- VIX effects are not earnings-specific

INTERPRETATION

- Evidence of state dependence
- Robust across timing decompositions

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Main Findings Summary

Hypothesis	Conclusion	Key Stats
H1	PreSkew predicts stronger normalization	$\beta \approx 0.63,\ t \approx 53$
H2 (Main)	High VIX weakens immediate resolution	$\beta \approx -0.16,\ t \approx -14$
H2 (Timing)	High VIX delays adjustment to later days	UR1: -0.169; UR23: $+0.122$

Summary

Earnings announcements reshape both the structure and level of uncertainty; macro fear weakens and delays this process.

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Contributions and Implications

Theoretical

- Extends study from IV levels to IV structure (skew)
- Integrates downside beliefs into repricing

Empirical

- Documents systematic skew normalization
- Identifies macro-state dependence in resolution

Practical

- Traders: high VIX delays vol compression
- Analysts: skew helps predict pricing corrections

Bottom Line

Announcements do not merely reduce implied volatility; they reshape downside-tail pricing in a state-dependent manner driven by macro uncertainty.

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Conclusion

- **Beyond volatility level:** Earnings announcements do more than reduce *how much* uncertainty – they also reshape *what kind* of uncertainty (asymmetric tail risk).
- **State-dependent disclosure:** The market's response to firm-specific news is not uniform: high aggregate fear (VIX) weakens and delays uncertainty resolution.
- **Two dimensions of repricing:** The option market corrects both the **structure** of downside beliefs (skew normalization) and the **level** of expected volatility (IV compression), but these adjustments operate at different speeds and depend on pre-event conditions.

Main Message

Earnings announcements reshape both the **level** and the **structure** of option-implied uncertainty.

Future Research

Limitations (honest caveats)

- Causal identification incomplete – pre-skew may embed information leakage or mean reversion
- H2 weaker in Fama-MacBeth (VIX has limited cross-sectional variation)
- Mechanisms not distinguished: attention vs. risk premium
- No pseudo-event test to rule out pure mean reversion of skew

Future research (where to go)

- **Test mechanisms:** Interact VIX with institutional ownership / analyst coverage
- **Exogenous shocks:** Use political events, macro surprises as instruments for VIX
- **Cumulative UR:** Longer-window analysis to confirm *delayed* (not reduced) resolution
- **Pseudo-event:** Compare skew reversal around earnings vs. random dates

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Thank You!

Questions and comments are welcome

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Predicting the Closing Cross

Chen, Kwok, Shi & Gang (2026)

What This Research Does & Does Well

- Core claim: closing auction-driven price pressure, captured by DriftxAuc = ClosingDrift $\times$ AucShare, predicts **overnight mean reversion** via non-informational flow.

Key Contributions

- Scales Bogousslavsky & Muravyev (2020) from DJIA-30 to full S&P 500
- Introduces **interaction signal** DriftxAuc, outperforming all 4 alternatives
- Quantifies **2.23**$\times$ **coefficient amplification** in volatile regimes via Markov switching
- Addresses the market-beta confound via rolling beta adjustment (ΔSharpe: 3.58 $\rightarrow$ 4.24)

Concern 1: Look-Ahead Bias in Signal Construction

NYSE Intraday Timeline

~3:45 PM
NYSE publishes OII:
imbalance direction & size

3:58 PM
Deadline to submit or
cancel MOC orders

4:00 PM
Continuous session halts;
closing auction runs

4:00:XX PM
P_{close} published;
Signal now computable;
Entry window: CLOSED

Look-Ahead Problem in Primary Interactive Signal

$$\text{ClosingDrift} = \frac{P_{\text{close}} - P_{\text{4pm}}}{P_{\text{4pm}}}, \quad \text{AucShare} = \frac{\text{CSize}}{\text{TotalVol}}$$

- P_{close} is the auction clearing price, which does not exist until after the auction has closed.
- AucShare cannot be finalized until post-close volume is confirmed.
- **DriftxAuc is unobservable until after the entry window is gone.** The portfolio enters position at P_{close}, a direct look-ahead bias.

How Much Is This Bias?

Only observable *pre-auction* signal is M4 (signed order imbalance):

Signal	OLS t	Sharpe
M1: DriftxAuc *(post-auction)*	-9.97^{***}	3.33
M4: Signed imbalance *(pre-auction)*	-3.65^{***}	0.77

, which inadvertently benchmarks its own bias: the 4.3× Sharpe gap between M1 and M4 ~ the cost of removing the look-ahead.

Concern 1 (Cont'd): Realistic Entry Destroys the Economics

Paper's World vs. Reality

	Paper assumes	Reality
Entry market	Regular session	After-hours ECN only
Entry spread	∼1 bps	**5–15 bps**
Entry liquidity	Full book	Thin, no mkt makers
Exit market	Next open	Next open auction
Exit spread	Assumed low	Wide; opening auc.

Entry friction alone might exceed the entire gross return

What a realistic backtest requires:

1. **Entry at after-hours mid:** first ECN quote after P_{close}; add half-spread of 5–8 bps.
2. **Exit at open mid:** opening auction print plus 3–5 bps for illiquid names in Q1/Q5.
3. **Net return likely unprofitable:** $4.58 - 5.0_{\text{entry}} - 4.0_{\text{exit}} \approx -4.4$ bps

The Only Viable Implementation

- Ingest NYSE OII feed at 3:45, 3:50, 3:55 PM
- Run cross-sectional regression predicting ClosingDrift from imbalance size, direction, and recent price momentum
- Rank names by predicted DriftxAuc analog
- Submit **MOC orders before 3:58 PM** cutoff for top and bottom deciles
- Exit via limit orders through opening auction and first 30 minutes of continuous trading

This is the only version without look-ahead bias: entry executes at P_{close} via the auction itself, with zero market impact by construction. The forecast problem is harder — M4 suggests Sharpe ≈ 0.77 gross.

Concern 2: The FM t-stat Reflects Days You Cannot Trade

Selected Regression Result

Estimator	S&P 500	DJIA
Pooled OLS t	-2.70^{***}	-4.48^{***}
Fama–MacBeth t	-10.55^{***}	-4.30^{***}
Ratio	**3.9**×	1.0×

FM runs 1,885 daily cross-sections and tests whether the mean slope $\bar{b}$ differs from zero. It weights every day equally, regardless of whether markets were calm or in crisis.

Why Volatile Days Dominate $\bar{b}$

- The 310 volatile-regime days produce slopes 2.23× larger than the 1,701 calm days: $\beta_{\text{vol}} = -1.392$ vs. $\beta_{\text{calm}} = -0.624$
- Larger slopes on volatile days pull $\bar{b}$ away from zero because those days have far greater cross-sectional dispersion in DriftxAuc.

The Practical Point

- Volatile-regime days (COVID, 2022 Fed tightening): after-hours spreads are *widest*, short-borrow costs *spike*, and execution risk is *highest*.
- The days that make it strong are the days a fund **cannot** trade it. *Subsample regression on calm days only would tell the implementable Sharpe.*

Minor Points & Future Work

Minor Clarifications Needed

- **Survivorship bias:** S&P 500 constituents rebalance over time; current-period tickers may overstate performance. A matched predecessor file (CRSP delist codes) would address this.
- **Earnings nights:** the $|ret| > 3\%$ filter removes large *intraday* moves but misses overnight gaps driven by after-market earnings surprises. An explicit I/B/E/S exclusion would tighten the mechanism claim.
- **Markov probs:** smoothed probs use future data — not tradeable in real-time. Re-run regime tests with *filtered* Hamilton (1989) probs before claiming live applicability.

Interesting Extensions

- **Nasdaq closing cross:** Nasdaq runs a parallel auction at 4:00 PM. Testing DriftxAuc there distinguishes NYSE microstructure from a universal equity phenomenon.
- **Pre-close imbalance update:** NYSE disseminates Order Imbalance Information from 3:45 PM. Incorporating real-time feeds could enable an intraday signal update.
- **Connection to our paper:** our H2 shows high VIX *dampens* IV compression around earnings; here, high volatility *amplifies* overnight reversals by $2.23\times$. Both indicate macro volatility reshapes how prices respond to scheduled market events.

Overall Assessment

Strengths

- **Clean economic mechanism** with a clear null: non-informational flow $\Rightarrow$ temporary dislocation $\Rightarrow$ mean reversion
- **Comprehensive robustness:** event filters, sub-period splits, 5 signal alternatives, beta adjustment comparison
- **State-dependent insight** via Markov switching goes beyond a single-regime story

Three Actionable Asks

1. **Re-run backtest with after-hours entry price**: record first ECN mid-quote after P_{close} is disseminated; report net Sharpe under 5, 8, 10 bps one-way entry cost
2. **Report FM t-stat on calm days only** (1,701 obs): decompose whether the signal survives outside crisis/volatile episodes where execution is hardest
3. **Pre-auction signal test**: build DriftxAuc equivalent using 3:45–3:58 PM OII data only, enter via MOC orders; this is the only version that eliminates the look-ahead bias